

Pi Zero 2W Kiosk Setup Guide

DietPi + Wayland + Cage + Cog

Hardware-accelerated portrait kiosk for the Family Dashboard.
Resolves icon rendering, mouse cursor, and Wayland permission issues.

System Requirements

- Hardware: Raspberry Pi Zero 2W + HDMI display (portrait)
- OS: DietPi (Debian Bookworm)
- Target: Web dashboard URL (e.g. <http://192.168.102.230:8080/kiosk>)

Step 1 - Install Dependencies

Install the Wayland compositor, WebKit browser, and essential font packages.

```
sudo apt update
sudo apt install -y cage cog wlr-randr fonts-noto-color-emoji fontconfig
```

fonts-noto-color-emoji prevents icon glyphs from showing as empty squares (tofu).

Step 2 - Kernel-Level Cursor Removal

Permanently hides the mouse cursor at the kernel level (no flicker, no workarounds).

```
echo 'ENV{ID_INPUT_POINTINGSTICK}=""
ENV{ID_INPUT_MOUSE}=""' | sudo tee /etc/udev/rules.d/99-hide-cursor.rules
sudo udevadm control --reload-rules && sudo udevadm trigger
```

Step 3 - Add User to Required Groups

The kiosk user needs video, input, and render permissions for GPU and display access.

```
sudo usermod -a -G video,input,render dietpi
```

Changes take effect after logout/login or reboot.

Step 4 - Create Startup Script

Create /home/dietpi/start-kiosk.sh - replaces YOUR_DASHBOARD_URL with your actual URL.

```
#!/bin/bash
sleep 1
wlr-randr --output HDMI-A-1 --transform 90
export WEBKIT_CHOICE_MIXED_CONTENT_POLICY=0
export WEBKIT_DISABLE_WEB_SECURITY=1
exec /usr/bin/cog "http://YOUR_DASHBOARD_URL_HERE"
```

```
chmod +x /home/dietpi/start-kiosk.sh
```

Step 5 - Create Systemd Service

Create /etc/systemd/system/kiosk.service - note: no WAYLAND_DISPLAY in Environment.

```
[Unit]
Description=Kiosk Browser

[Service]
Type=simple
User=dietpi
Environment=XDG_RUNTIME_DIR=/run/user/1000
ExecStart=/usr/bin/cage /home/dietpi/start-kiosk.sh
Restart=always
RestartSec=5

[Install]
WantedBy=multi-user.target
```

Step 6 - Enable Linger & Create Runtime Dir

Ensure `/run/user/1000` exists so Cage can create its Wayland socket.

```
sudo loginctl enable-linger dietpi
sudo mkdir -p /run/user/1000
sudo chown dietpi:dietpi /run/user/1000
```

Step 7 - Final Activation

```
sudo fc-cache -fv
sudo systemctl daemon-reload
sudo systemctl enable --now kiosk.service
```

The display should go blank briefly and then show the dashboard in fullscreen portrait mode.

Troubleshooting

Cage cannot open Wayland socket

Fix: `XDG_RUNTIME_DIR` missing or wrong. Run Step 6 (enable-linger, mkdir, chown).

DRM session / Permission denied

Fix: User not in video/input groups. Run Step 3 and reboot.

EGL / DRI2 / renderD128 permission denied

Fix: User not in render group. Run Step 3 and reboot.

Blank screen / Cage exits immediately

Fix: Check the URL in `start-kiosk.sh`. Verify cage dependency: ensure no `WAYLAND_DISPLAY` is set in the service Environment.

Screen stays on after reboot

Fix: Check `journalctl -u kiosk.service --no-pager` for errors.